

Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

Michał Pytel¹, Paweł Pozorski¹

¹ Warsaw University of Technology, Poland

{michal.pytel2, pawel.pozorski}@pw.edu.pl

Abstract

Between 70–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model of Baltic and North Sea traffic. Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and a COLREGs-style give-way rule (International Regulations for Preventing Collisions at Sea). The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses DTU SIMCOL [2]; route-network collision frequency is checked with IWRAP Mk II [3]; rescue follows IAMSAR practice [4] with MASSIM/Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. We compare a classical baseline, a weather-aware baseline, and a fatigue-aware Proposed System across four scenarios (30 seeds each; 360 runs) with Mann–Whitney U tests, Holm–Bonferroni correction, and Cliff’s δ . Pre-registered fatal-rate and survival thresholds (H1–H3) were not confirmed after correction. Preparedness P_{prep} rose with large confirmed effects in every scenario ($|\delta| \geq 0.90$). Mean Storm Corridor collisions fell by about 28 % versus Baseline A ($\delta=0.32$, not significant after Holm).

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors. The International Maritime Organization (IMO)¹ attributes 70–96 % of incidents to human causes [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters. Hull and machinery claims remain large [9].

The International Regulations for Preventing Collisions at Sea (COLREGs) [10]² assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime ABMs treat only loosely [11, 12, 13]:

- **Weather and radio.** Sea clutter and storms reduce the

chance that a marine VHF³ warning is heard and acted on. Many models assume that once risk is detected, the manoeuvre always happens.

- **Crew fatigue.** Fatigue builds over a watch and follows a circadian pattern [14]. Prior work often keeps fatigue in a side model, without a direct link to whether a warning is executed.

This paper couples those effects in one product: warning success depends on both local weather and crew fatigue. The model runs on real Baltic–North Sea geography (Section 3). Routes are synthesised from bathymetry and Automatic Identification System (AIS)⁴ traffic density (Section 3.3). Avoidance uses Closest Point of Approach / Time to CPA (CPA/TCPA)⁵ and a simple give-way rule (Section 4). Damage uses the DTU ship-collision model SIMCOL [2] (Section 5); frequency is checked with IALA IWRAP [3] (Section 7); survivors enter a search-and-rescue (SAR)⁶ chain (Section 6). The engine is Rust on krABMaga [15]. Figure 1 summarises the causal chain.

2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability reduces fatal outcomes more than classical, weather-unaware practice. We implement three configurations:

Baseline A (Classical) COLREGs give-way, no weather-aware slowdown, unmanaged fatigue. Storm radio loss still applies inside the storm zone — it is environmental, not a privilege of other methods.

Baseline B (Shore Rest Alerts) Weather-aware speed reduction and slower fatigue build-up; no further mitigation.

Proposed System Smart watch scheduling (lowest fatigue build-rate) on top of Baseline B. Six KPIs are defined in Table 3.

We use them for the following hypotheses:

- H1.** Proposed reduces the fatal-event rate (Table 3, KPI 1) by $\geq 20\%$ versus Baseline A ($p < 0.05$, Mann–Whitney U, Cliff’s $|\delta| \geq 0.2$).
- H2.** Proposed improves post-incident survival (KPI 4) by $\geq 15\%$ versus Baseline A.
- H3.** Under Storm Corridor, Proposed beats Baseline B on the fatal-event rate ($p < 0.05$).

³https://en.wikipedia.org/wiki/Marine_VHF_radio

⁴https://en.wikipedia.org/wiki/Automatic_identification_system

⁵Standard radar/AIS encounter metrics: distance and time to the predicted closest point of approach. Overview: https://en.wikipedia.org/wiki/Collision_avoidance

⁶https://en.wikipedia.org/wiki/Search_and_rescue

¹https://en.wikipedia.org/wiki/International_Maritime_Organization

²https://en.wikipedia.org/wiki/International_Regulations_for_Preventing_Collisions_at_Sea

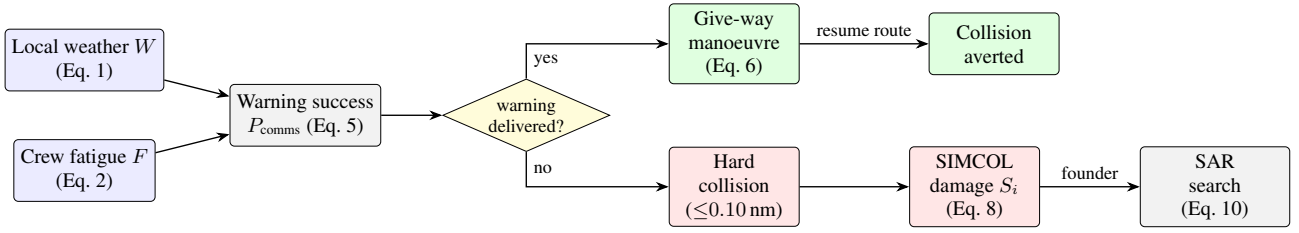


Figure 1: *Causal pipeline.* Weather W and fatigue F set warning success P_{comms} (Eq. 5). A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions are scored with SIMCOL; a foundering ship enters SAR.

3. Simulation Model

This section defines the world, agents, routes, and weather field used by all methods.

3.1. Domain and Time

The domain is the Baltic and North Sea: latitude $\phi \in [50.5^\circ, 66.0^\circ]$ N and longitude $\lambda \in [-5.0^\circ, 31.0^\circ]$ E. An equirectangular⁷ map with a mid-latitude correction sets one field unit to one nautical mile⁸ at $\phi_0 = 58.25^\circ$:

$$x = (\lambda - \lambda_{\min}) k_{\text{lon}}, \quad y = (\phi - \phi_{\min}) k_{\text{lat}},$$

with $k_{\text{lat}} = 60 \text{ nm}^\circ$ and $k_{\text{lon}} = 60 \cos \phi_0$. East–west distortion at the box edges is about 6–10 % and is left uncorrected; distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth⁹ sit in an R-tree¹⁰ for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3).

Time steps are $\Delta t = 15 \text{ min}$ (0.25 h). The default run length is $T = 2880$ ticks (30 days). A wall-clock variable starts at 06:00 UTC and drives the circadian fatigue term (Section 4). Each run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship i has position $\mathbf{p}_i$, waypoint route R_i , travel direction $d_i \in \{+1, -1\}$, heading ψ_i , cruise speed v_i , state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew $n_i = 10$, and fatigue $F_i \in [0, 1]$.

Routes are synthesised once (Section 3.3) and stored as `data/ais.paths.json`. Ports are polygons in `data/ports.json`. Entering a port docks the ship. Cruise speed is drawn at spawn ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

Waypoint pursuit. The active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within 0.5 nm. The step length is $s = \min(v_i \Delta t, \rho)$. Moves use up to five geometrically halved

lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses. Fleet size N is held fixed: when a ship is removed, a new one is spawned on a random synthesised track.

Same-destination following. Co-directional ships with end-points within 5 nm share a destination. A trailer within 3 nm of a leader matches the leader’s speed; within 0.5 nm it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the EMODnet DTM [16]. Lane attractiveness comes from EMODnet vessel-density maps [17, 12]. Both are fetched once via the OGC Web Coverage Service (WCS)¹¹ and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water.

Each class c has draft T_c and under-keel clearance (UKC _{c})¹² [18]. A cell of depth h is navigable iff $h \geq T_c + \text{UKC}_c + m_s$ with buffer $m_s = 2 \text{ m}$ (Table 1). Deep-draft very large crude carriers (VLCCs)¹³ are excluded (Danish Straits).

Table 1: *Minimum transit depth by vessel class (draft + under-keel clearance).*

Class	Draft T_c (m)	UKC _{c} (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

Between sampled port pairs, a weighted A* planner [19] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h s_h + w_d s_d + w_\ell (1 - a_\ell),$$

penalising shallow margins (s_h), proximity to shore/shoal (s_d), and off-lane cells (a_ℓ). Default weights are $(w_h, w_d, w_\ell) = (3, 4, 2.5)$. Paths are simplified under a depth constraint and checked against raw bathymetry every 0.15 nm. Seeded cost jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

¹¹https://en.wikipedia.org/wiki/Web_Coverage_Service

¹²https://en.wikipedia.org/wiki/Under_keel_clearance

¹³https://en.wikipedia.org/wiki/Very_large_crude_carrier

⁷https://en.wikipedia.org/wiki/Equirectangular_projection

⁸https://en.wikipedia.org/wiki/Nautical_mile

⁹https://en.wikipedia.org/wiki/Natural_Earth

¹⁰<https://en.wikipedia.org/wiki/R-tree>

3.4. Weather Field

A 50×50 grid holds sea state s , visibility v , and wind u in $[0, 1]$. They fuse into hazard $W_{ij} \in [0, 1]$:

$$W = \text{clip} \left(\frac{w_s s + w_v (1 - v) + w_u u}{w_s + w_v + w_u}, 0, 1 \right), \quad (1)$$

with $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, AR(1)¹⁴ background fields, overlays, coupling, and smoothing (parameters in Appendix A).

Independently, Storm Corridor places a moving storm of radius r_{storm} that drifts at 0.30 nm/tick along a fixed track (English Channel → North Sea → Skagerrak → Kattegat → Baltic → Gdańsk). Inside the zone, communication and (for non-A methods) speed are degraded.

4. Crew Fatigue and Collision Avoidance

Fatigue and avoidance form the mechanism behind H1–H3: warnings fail when weather is bad and crews are tired.

4.1. Fatigue Dynamics

Fatigue $F_i \in [0, 1]$ updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \frac{1}{2} \left(1 - \cos \left(\frac{2\pi(t_{\text{utc}} - 3)}{24} \right) \right) \in [0, 1].$$

While ACTIVE, with local hazard W ,

$$\Delta F = (\beta_{\text{base}} + \beta_W W + \beta_c c(t_{\text{utc}})) m(\text{method}), \quad (2)$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015, 0.0040, 0.0008)$ and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \quad (3)$$

then $F_i \leftarrow \min(1, F_i + \Delta F)$. While DOKED, recovery is 0.025 per tick.

Above threshold $F^* = 0.40$, fatigue reduces warning reliability with maximum penalty $\kappa = 0.65$:

$$g(F_i) = \begin{cases} 1, & F_i \leq F^* \\ 1 - \kappa \frac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35, 1]. \quad (4)$$

4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position $\Delta \mathbf{p}$ and relative velocity $\Delta \mathbf{v}$,

$$t^* = - \frac{\Delta \mathbf{p} \cdot \Delta \mathbf{v}}{\|\Delta \mathbf{v}\|^2}, \quad d_{\text{CPA}} = \|\Delta \mathbf{p} + t^* \Delta \mathbf{v}\|,$$

where t^* is in ticks. A manoeuvre is considered only if $\|\Delta \mathbf{p}\| \leq 20$ nm, $d_{\text{CPA}} \leq 5$ nm, $0 < t^* \leq 12$ ticks, and the give-way ship is not in cooldown.

¹⁴https://en.wikipedia.org/wiki/Autoregressive_model

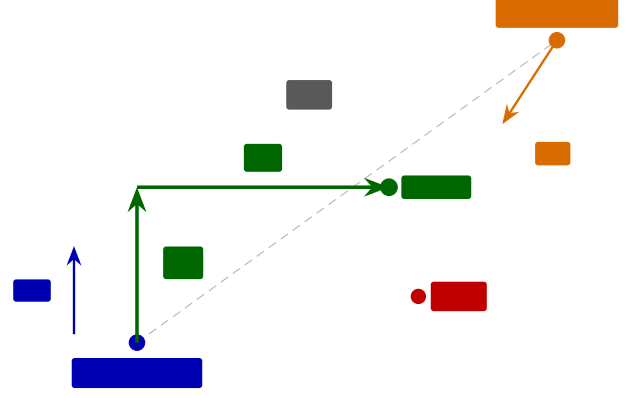


Figure 2: Give-way geometry. Ship a steers to a temporary waypoint $\mathbf{w}_{\text{avoid}}$ placed L_f ahead and L_o to starboard (Eq. 6). The manoeuvre runs only if the warning is delivered (Eq. 5).

The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min(q(\mathbf{p}_a), q(\mathbf{p}_b))}_{\text{weather / storm}} \cdot \underbrace{\min(g(F_a), g(F_b))}_{\text{fatigue (Eq. 4)}}, \quad (5)$$

where $q(\mathbf{p})$ is nominal link reliability, reduced to the storm-comms rate (0.08 inside Storm Corridor) when $\mathbf{p}$ is in the storm zone. A uniform draw below P_{comms} executes the manoeuvre; otherwise geometry alone decides the encounter. Equation 5 is the central coupling of the model.

A give-way ship receives one temporary waypoint $L_f = 3$ nm ahead and $L_o = 5$ nm to starboard of heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f (\sin \psi, \cos \psi) + L_o (\cos \psi, -\sin \psi). \quad (6)$$

A short message log records the warning exchange. A 3-tick cooldown prevents thrashing.

4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [20]:

$$\mathbf{F}_{\text{rep}} = \sum_{k: d_k < d_o} \frac{k_{\text{rep}}}{d_k^p} \hat{\mathbf{n}}_k,$$

with exponent $p = 2$ by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm (≈ 185 m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS)¹⁵. Damage uses DTU SIMCOL (ship collision damage / survivability) [2].

¹⁵https://en.wikipedia.org/wiki/Vessel_traffic_service

Absorbed energy follows Pedersen and Zhang [21, 2] on the normal closing component:

$$E_{\text{abs}} = \frac{1}{2} \frac{M'_a M'_b}{M'_a + M'_b} V_n^2, \quad V_n = |(\mathbf{v}_b - \mathbf{v}_a) \cdot \hat{\mathbf{n}}|, \quad (7)$$

with added-mass factors $m_{\text{sway}} = 0.85$ and $m_{\text{surge}} = 0.05$ [2]. Minorsky's correlation [22] links energy to damaged volume. Relative penetration t/B drives a survival factor shaped like SOLAS Chapter II-1 probabilistic damage stability [23]¹⁶:

$$S_i = \begin{cases} 1, & t/B \leq r_0 \\ \left(\frac{r_1 - t/B}{r_1 - r_0} \right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \geq r_1, \end{cases} \quad (8)$$

with $r_0 = 0.10$, $r_1 = 0.30$. This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b (1 - S_i), \quad (9)$$

drawn as independent Bernoulli trials. If $S_i < S^+ = 0.50$, the struck ship founders and the whole crew enters the liferaft (EVAC). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3).

Encounter type is diagnostic, from the folded heading angle θ :

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45^\circ \\ \text{front-to-side (crossing)}, & 45^\circ \leq \theta < 135^\circ \\ \text{head-on}, & \theta \geq 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

6. Search and Rescue

Foundering ($S_i < S^+$) starts the liferaft chain, following International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual practice [4]¹⁷. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). A second chain handles persons overboard from afloat ships (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width w , search speed v_s , and area A ,

$$\text{CDP} = 1 - \exp\left(-\frac{n w v_s \tau}{A}\right) = 1 - e^{-C}. \quad (10)$$

Search area grows as $A(\tau) = \pi(r_0 + u_d \tau)^2$ with initial radius r_0 and drift u_d . Each on-scene tick draws Bernoulli detection at coverage $1 - e^{-dC}$.

¹⁶https://en.wikipedia.org/wiki/International_Convention_for_the_Safety_of_Life_at_Sea

¹⁷https://en.wikipedia.org/wiki/International_Aeronautical_and_Maritime_Search_and_Rescue_Manual

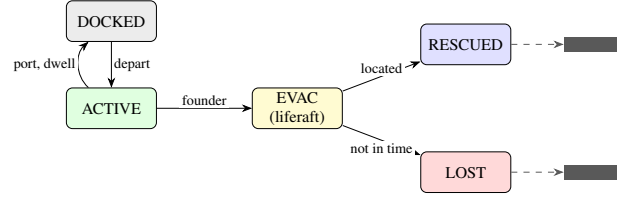


Figure 3: *Vessel state machine. Foundering collision → EVAC; SAR yields RESCUED (Eq. 10) or LOST. Terminal ships respawn to keep fleet size. KPI fatalities count only unrecovered persons-in-water (Section 6.3).*

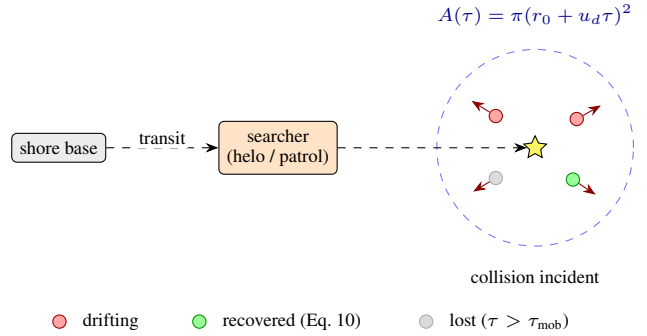


Figure 4: *Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person. Unrecovered people within τ_{mob} are the fatal KPI events.*

6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\text{wci} = 13.12 + 0.6215 T - 11.37 U^{0.16} + 0.3965 T U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately 1.00/0.95/0.77/0.60 (helicopter) and 1.00/0.80/0.40/0.21 (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within $\tau_{\text{mob}} = 12$ ticks (≈ 3 h) is lost. **This is the only path that increments the fatal KPI.** Liferaft LOST is logged but does not add to fatal_per_1k_hrs. Liferaft endurance is 96 ticks (24 h).

7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. To place traffic intensity on a known scale, the committed route network is analysed with IALA's IWRAP Mk II waterway risk model [3, 24]¹⁸:

$$N_c = N_a P_c.$$

¹⁸https://en.wikipedia.org/wiki/International_Association_of_Marine_Aids_to_Navigation_and_Lighthouse_Authorities

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows Q , beams B , speeds V , Pedersen diameter for crossings [3, 25]). Causation probabilities are the IWRAP defaults: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$.

Each batch row stamps N_c from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the present network, N_c is on the order of 0.01 collisions/yr, inside the usual <1 /yr/fairway acceptance band.

8. Experimental Design

8.1. Methods and Scenarios

The three methods of Section 2 differ as follows:

- **Baseline A:** no weather-aware slowdown (storm speed factor = 1); unmanaged fatigue ($m = 1.00$).
- **Baseline B:** hazard and storm slowdowns; $m = 0.82$.
- **Proposed:** same slowdowns as B; $m = 0.65$.

SIMCOL parameters are identical across methods. Storm-zone radio loss applies to all methods.

Table 2 lists the four scenarios. Each condition uses 30 seeds and 2880 ticks ($4 \times 3 \times 30 = 360$ runs).

8.2. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add 0.25 h per ACTIVE ship per tick. A *fatality* in the KPIs is an unrecovered person-in-water (Section 6.3).

Crew preparedness uses method awareness $\alpha_M \in \{2.0 \text{ (A)}, 1.3 \text{ (B)}, 1.0 \text{ (Proposed)}\}$:

$$P_{\text{prep}} = \max(0, 1 - \alpha_M W) \cdot \max(0, 1 - 0.70 F).$$

Diagnostic counters (collision-type mix, MOB tallies) are logged but not entered into the Holm battery.

8.3. Statistical Analysis

For each (KPI $\times$ scenario $\times$ method pair) we use a two-tailed Mann–Whitney U test¹⁹ with Cliff’s δ [26]. Multiplicity uses Holm–Bonferroni²⁰ step-down over the full battery. A result is *significant* if $p_{\text{corr}} < 0.05$ and *confirmed* if also $|\delta| \geq 0.2$. Pairs are (A vs Proposed), (B vs Proposed), and (A vs B). Seeds are matched across conditions.

9. Implementation

The engine is Rust on krABMaga [15]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE)²¹. Each finished run writes a JSONL tick log (every ten ticks), a manifest, and a SQLite²² results row. A React workbench (deck.gl / MapLibre) replays completed runs.

¹⁹https://en.wikipedia.org/wiki/Mann-Whitney_U_test

²⁰https://en.wikipedia.org/wiki/Holm-Bonferroni_method

²¹https://en.wikipedia.org/wiki/Server-sent_events

²²<https://en.wikipedia.org/wiki/SQLite>

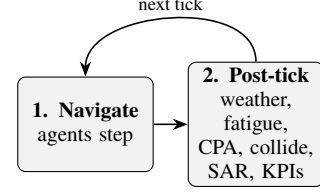


Figure 5: *Macro-tick order. Agents navigate, then the post-tick pipeline updates weather, human factors, encounters, and outcomes.*

9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline shown in Fig. 5: weather advance; grounding revert; hazard and storm speed damping; fatigue; CPA avoidance and comms draws; following; rising-edge collisions and SIMCOL; SAR/MOB; fleet top-up; KPI accumulate; optional JSONL frame. Navigation therefore uses weather and avoidance state from the previous tick’s post-tick pass.

9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max(0.30, 1 - 0.45 W).$$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

10. Results

The definitive batch ran 30 seeds $\times$ 4 scenarios $\times$ 3 methods (360 runs). Confirmation requires $p_{\text{corr}} < 0.05$ and $|\delta| \geq 0.2$. Figure 6 shows per-run KPI distributions; Figure 7 summarises effect sizes for the pre-registered contrasts.

10.1. Calibration and Storm Contrast

Under Baseline A, Calm Passage mean collisions were 0.60 per 1 000 ship-hours (inside the 0.5–1.5 band). Storm Corridor raised that rate to 1.48 ($2.48 \times$ calm), restoring a clear weather contrast after the depth+lane route rebase.

10.2. Hypotheses H1–H3

Table 4 summarises the pre-registered contrasts. **H1** and **H2** are not confirmed: fatal rates and survival ratios sit near floor/ceiling in Calm Passage, Storm Corridor, and Blind Shore, so tests have little power. Deep Water Rescue has a higher fatal base rate; Proposed reduces mean fatalities from 0.038 (Baseline A) to 0.023 ($\approx 39\%$), but survival is almost unchanged and neither contrast survives Holm ($p_{\text{corr}} = 1$, $|\delta| \leq 0.02$). **H3** is non-significant (means 0.0022 vs 0.0024).

10.3. Mechanism and Collisions

Mean P_{prep} is higher for Proposed than for either baseline in every scenario, with confirmed effects ($|\delta| \in [0.90, 1.00]$). Mean Storm Corridor collisions fall from 1.48 to 1.07 versus Base-

Table 2: *Scenario summary (30 seeds, 2880 ticks/run).*

Scenario	Key conditions	Ships	Hyp.
Calm Passage	Calm weather; calibration reference (target band ≈ 0.5 – 1.5 collisions / 1 k ship-hrs under Baseline A).	20	Calib.
Storm Corridor	Moving storm; storm-comms rate 0.08; non-A speed factor 0.45.	25	H1, H3
Blind Shore	Fleet-wide comms 0.55; tighter avoidance berth ($L_o = 3$ nm).	25	H1
Deep Water Rescue	Sparse fleet; SAR mobilisation 4 ticks; helo/patrol 55/16 kn.	15	H2

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	<code>fatal_per_1k_hrs</code>	Fatalities $\div$ ship-hrs $\times 10^3$	H1, H3
2	<code>collision_per_1k_hrs</code>	Collisions $\div$ ship-hrs $\times 10^3$	Calib.
3	<code>evac_activation_rate</code>	Evac activations $\div$ ship-hrs $\times 10^3$	—
4	<code>survival_ratio</code>	$1 - \text{fatalities} / \text{crew exposed in collisions}$	H2
5	<code>avg_tta_hours</code>	Mean rescue time-to-arrival	H2
6	<code>mean_p_prep</code>	Time-averaged crew preparedness	mech.

Table 4: *Pre-registered hypothesis outcomes (definitive 30-seed batch).*

Hyp.	Contrast	Verdict	Cliff’s δ
H1	Fatal rate, Proposed vs A	Not confirmed	$ \delta \leq 0.07$
H2	Survival ratio, Proposed vs A	Not confirmed	$ \delta \leq 0.07$
H3	Fatals under Storm, Proposed vs B	Not confirmed	$\delta \approx 0.00$

line A ($\approx 28\%$, $\delta=0.32$), but that contrast is not significant after family-wise Holm correction.

11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. Methods differ by fatigue build-rate m (Eq. 3), weather/storm speed awareness, and preparedness weights α_M . The batch confirms the preparedness lever and a directional storm-collision benefit, but not the pre-registered fatal/survival thresholds. Fatal events are rare outside Deep Water Rescue, so H1–H3 are under-powered even when point estimates move in the expected direction.

Accredited layers add external structure: SIMCOL for damage [2], IWRAP for network frequency [3], and MAS-SIM/Ashrafi for rescue [5, 7]. The survival factor S_i (Eq. 8) is a SOLAS-shaped surrogate, not a full residual-stability calculation.

Main limitations: synthesised routes trade some traffic realism for control; the lower-id give-way rule simplifies COLREGs; force-field avoidance is off by default; KPI fatalities count only MOB timeouts; IWRAP stamps use static routes rather than per-run tracks. Calibration under Baseline A Calm Passage (0.60 collisions per 1000 ship-hours) sits near published AIS near-miss bands [12, 13]. Longer runs or collision-rate primary endpoints would raise power on fatal contrasts.

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Definitive batch KPIs (30 seeds & 4 scenarios × 3 methods)

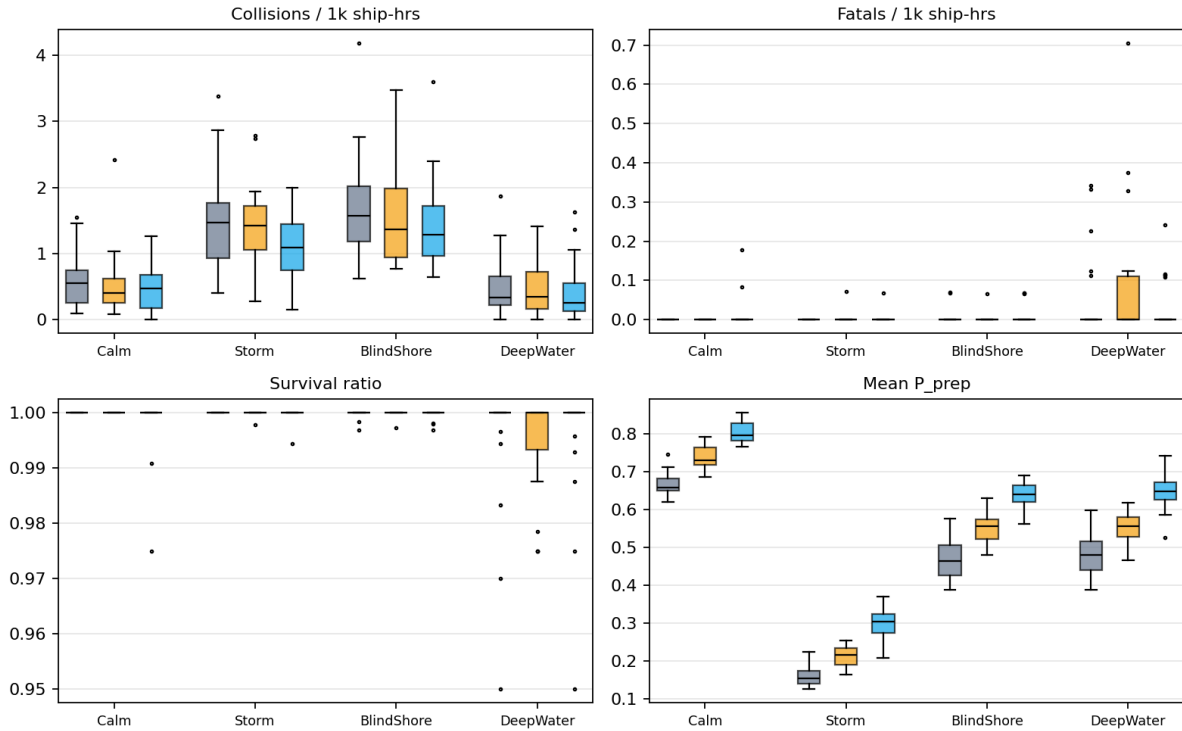


Figure 6: Per-run KPI distributions for the definitive batch (30 seeds). A = Baseline A, B = Baseline B, P = Proposed.

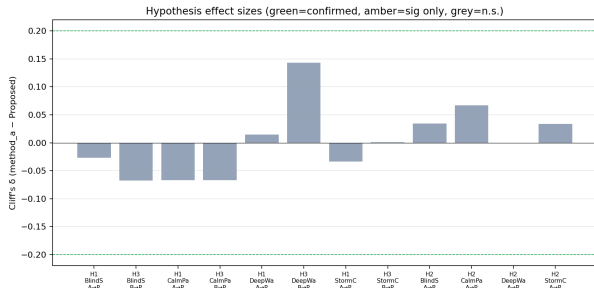


Figure 7: Cliff's δ for H1-H3-relevant pairs. Green: confirmed; amber: significant only; grey: not significant after Holm.

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Table 5: Selected engine default constants.

$(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$; gradient cap 0.12. Calm and Stormy presets scale the flip rates.

Constant	Default	Role
Δt	0.25 h	tick length (15 min)
T	2880	ticks per run (30 d)
weather grid	50×50	hazard field
(w_s, w_v, w_u)	(1.0, 0.5, 0.8)	hazard fusion weights
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o	3 / 5 nm	give-way waypoint offsets
cooldown	3 tk	re-manoeuve lockout
r_{coll}	0.05 nm	hit at $2r = 0.10$ nm
port collision exclusion	1 nm	harbour water
follow vicinity / gap	3 / 0.5 nm	speed-match / keep-distance
same-destination	5 nm	endpoint match radius
$(m_{\text{sway}}, m_{\text{surge}})$	0.85, 0.05	SIMCOL added mass
Minorsky a, b	47.2, 32.7	energy-volume (MJ, m ³)
$\ell / L_{pp}, \tau$	0.12, 0.08 m	damage length / side steel
$r_0, r_1, S^\dagger$	0.10, 0.30, 0.50	S_i knee / loss / founder
$(\beta_{\text{base}}, \beta_W, \beta_c)$	0.0015, 0.0040, 0.0008	fatigue rates
recovery	0.025/tk	dock fatigue recovery
F^*, κ	0.40, 0.65	comms-degradation knee
storm-comms (corridor)	0.08	Storm Corridor link rate
storm-comms (generic default)	0.30	when storm enabled outside corridor presets
n crew	10	per vessel
dwell	8–48 tk	port stay
helicopter / patrol	80 / 25 kn	rescue asset speeds
Δt_{mob}	2 tk	rescue mobilisation delay
helo / patrol cutoff	40 nm	SAR asset-selection range
w, r_0, u_d	5 nm, 2 nm, 0.2 nm/tk	MASSIM sweep / datum / drift
survival window	96 tk	liferaft endurance (24 h)
τ_{mob}	12 tk	MOB cold-water window (3 h)
MOB scatter	0.5 nm	person-in-water spread
P_c (IWRAP)	$0.5\text{--}1.3 \times 10^{-4}$	causation probability

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A. Default Constants

Table 5 lists principal defaults (overridable in configuration).

Weather-field process (Mixed preset): regime flips $p_{cs} = 0.03$, $p_{sc} = 0.08$; Calm/Storm reversion targets (0.30, 0.80, 0.25) / (0.72, 0.30, 0.75); up to 8 storm cells; AR(1) coefficients $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$,